

Chapter 159 — Active Surveillance of Prostate Cancer

IF YOU READ NOTHING ELSE

- Grade group (GG) 1 (Gleason 3+3=6) → active surveillance (AS): guideline standard for most
- GG1 metastatic potential ≈ zero; metastases = missed higher grade
- Misattribution ~30% (systematic biopsy) >> true progression 1–2%/yr
- Intervention ~25% at 5 yr, whatever the inclusion criteria
- 15-yr prostate cancer mortality (PCM) 0.5% (Hopkins) vs 5% (Toronto)
- Epstein: ≤2 cores, no core >50%, PSA density (PSAD) <0.15
- Confirm biopsy 1–2 yr; PSA 6-monthly; biopsy every 2–5 yr
- GG ≥3 → treat; volume or PSA rise → investigate, don't treat
- AS = deferred cure (life expectancy ≥10 yr); watchful waiting (WW) = palliation
- BRCA2 → consider definitive therapy, even if low grade

OVERDIAGNOSIS AND OVERTREATMENT

- Prostate overdiagnosis 87–94% if the whole reservoir were detected
- US Preventive Services Task Force (USPSTF): 2012 Grade D → 2018 Grade C, thanks to AS uptake
- AS → WW when life expectancy <5–10 yr (≈80 yr, earlier if comorbid)
- WW: no biopsy, no imaging; treat symptoms only
- Within 5 yr, only very aggressive high-grade disease kills
- Older WW trial outcomes ≠ modern AS

CLINICALLY INSIGNIFICANT CANCER

- Autopsy prevalence ≈ age as % (≥80% over 80)
- Lifetime prostate cancer death risk 3% (North America)
- PSA + systematic biopsy → cancer in 15–25%
- Harbour : death ≈ 20:1; diagnosis : death ≈ 6:1
- Gleason grade group = most powerful predictor
- GG2 = 3+4; GG3 = 4+3; GG4 = 8; GG5 = 9–10
- Pattern 3 molecularly near-normal; patterns 4/5 aberrant
- PTEN: present 90% (G3) vs deleted 70–90% (G4)
- PTEN loss alone ≠ metastasis; needs MYC amplification
- GG1 invades locally (extraprostatic extension, rarely seminal vesicles) → still "cancer"
- True progression 10–20% after 10 yr → lifelong follow-up
- Undergraded ~30%: 85% 3+4, 10% 4+3, 5% ≥8
- Grade progression ≠ automatic treatment (3+4 at 80 after AS from 65)
- 2005 Gleason revision → stage migration

Trials:

- SPCG-4 (Scandinavian, pre-PSA): PCM 19.6% vs 31.3%, relative risk 0.55, +2.9 life-years; benefit <65 yr and GG ≥3 only; 4+3 >5× risk of 3+4
- PIVOT (Veterans Affairs): hazard ratio 0.84; ~1 yr gained; low-risk observation PCM 2.7%
- ProtecT (UK, PSA-detected): no PCM or survival difference at 15 yr (all <3%); metastases 9.4% vs 4.5% vs 5.0%; 61% of monitoring arm treated
- ProtecT active monitoring ≠ AS: no scheduled biopsy
- ProtecT QoL: monitoring same or better in every domain; 24.4% alive untreated at 15 yr
- ASIST: MRI + targeted biopsy — no confirmation-biopsy difference; ~50% less AS failure at 2 yr

OUTCOMES OF ACTIVE SURVEILLANCE

- 10 prospective series (~10,000 men); GAP3 (Global Action Plan Prostate Cancer Active Surveillance): >15,000 men, 25 centres
- Commonest death on AS: cardiovascular; non-prostate death 10× prostate death

- Toronto: all Gleason 6 + selected intermediate risk; 50% eligible; 5% PCM (1.5% actual)
- Hopkins: Epstein only; 20% eligible; 0.5% PCM
- Baseline Gleason 7 on AS → ≥20% metastasis at 15 yr
- Very low risk = Epstein; low risk = GG1 + PSA <10
- High-volume Gleason 6 → still AS; add MRI, repeat biopsy or genomics
- Guidelines: NICE: MRI at enrolment; AUA (American Urological Association): MRI before confirmation biopsy
- NICE alone: intermediate risk → radical treatment or AS
- Neuroendocrine progression without PSA rise → check testosterone, image
- Occult high-grade predictors: PSAD, free/total PSA, cores, core %, age, race, restricted diffusion
- Misattributed upgrades mostly → GG2; beyond GG3 rare
- Intermediate risk on AS: ≥20% metastasis, 80% free at 15 yr

OTHER RISK FACTORS

- Black men: unequivocally AS candidates; consider more intense surveillance
- Family history inflation → relatives' low-grade disease still AS
- First-degree relative with metastases <60 yr → germline testing
- BRCA2 > BRCA1, ATM for upgrading; most BRCA2 upgraded by 10 yr
- Germline testing (National Comprehensive Cancer Network): metastatic/high risk; strong family history; consider if intraductal or cribriform
- Confirmation biopsy 1–2 yr (key points: 1–3 yr; can delay to yr 3)
- Digital rectal examination every 1–2 yr; MRI ≤ every 18–24 mo
- Operate if: GG ≥3 (unless life expectancy short)
- Don't operate for: Gleason 6 volume rise, PSA rise alone, small-volume GG2
- Compliance → better survival; outcomes assume compliance
- Diffusion-weighted imaging (DWI): diffusion inversely ∝ Gleason
- MRI before confirmation biopsy + systematic cores (NICE recommends)
- Serial MRI alone cannot exclude progression (negative predictive value 0.8–0.9)
- MRI-invisible cancer → more indolent
- ASIST site variation → MRI quality and reader experience matter
- Progression risk on AS 0.2–5% at 15 yr → modest biomarker gain
- Decipher: 22 genes — upgrading GG1, not GG2
- Oncotype DX Genomic Prostate Score: 12 genes, 4 pathways; mixed on AS

- **Polaris Cell Cycle Progression:** 31 + 15 housekeeping genes; CCP >2 → higher death

GENOMICS CAVEATS

- **Heterogeneity:** scores differ between foci in one gland
- **Bayes:** 3% prior, 90% specificity → 10% false positive (~3×)
- **Tissue genomics only** if high volume, high PSAD or GG2
- **Repeat genomic test adds nothing**
- **Genomics never sole basis for AS**

PSYCHOLOGY AND LIFESTYLE

- **10–15% elect treatment without indication**
- **Anxiety:** diagnosis, not AS; fare poorly: anxiety disorder, social isolation
- **Smoking cessation:** most cost-effective intervention in medicine
- **Two minutes of specialist counselling** can change smoking
- **Vigorous exercise** >3 h/wk (≥6× resting metabolic rate); **brisk walking** ≥7 h/wk
- **BMI ≥27.5 (Asian)** → adverse pathology
- **Each +5 kg/m² BMI** → higher prostate cancer mortality
- **Obesity affects later biopsies, not confirmation** → true progression
- **Men's Eating and Living (MEAL) randomised trial (vegetables):** no effect at 24 mo
- **Vitamin D 1000–2000 IU/day** (trial dose 4000 IU/day × 1 yr)
- **Selenium ≥140 µg/day** → higher mortality; **vitamin E (SELECT)** → more cancer

PHARMACOLOGIC INTERVENTION

- **REDEEM (dutasteride):** 38% less "progression", volume only; no less upgrading
- **5-alpha-reductase inhibitors:** not for AS prevention; fine if benign prostatic hyperplasia indication
- **REDUCE, PCPT (Prostate Cancer Prevention Trial):** slight high-grade concern with 5-alpha-reductase inhibitors
- **Antiandrogen "progression" benefit = cytorreduction;** only upgrading counts
- **Statins reasonable; metformin not routine** (MAST trial pending)

- **Apalutamide 240 mg × 90 d; enzalutamide 160 mg × 1 yr** — investigational

BARRIERS, NOMENCLATURE, ECONOMICS

- **Shared decision-making: 4 requisites;** physician = navigator, not driver
- **Biopsy of 24–38 transperineal cores** → erectile dysfunction risk
- **Self-discontinuation 7–15%**
- **IDLE = indolent lesion of epithelial origin;** "cancer" retained; GG 1/5 vs 6/10
- **Non-metastasising cancers:** basal cell carcinoma, gliomas
- **15-yr AS retention 43% (Toronto) / 55% (Hopkins)**
- **Biannual biopsy ≈ surgery cost by 10 yr**
- **Most cost-effective:** annual MRI, biopsy only if PI-RADS ≥4

CLASSIC TRAPS

- **Trap:** watchful waiting = AS — **Actually:** WW palliative, no biopsies
- **Trap:** SPCG-4 observation predicts AS outcome — **Actually:** pre-PSA, intermediate-high risk
- **Trap:** ProtecT excess metastases condemn AS — **Actually:** >½ higher risk, no protocol biopsies
- **Trap:** Gleason 6 metastasises — **Actually:** metastases = missed higher grade
- **Trap:** Very low vs low risk = metastatic risk — **Actually:** occult higher-grade likelihood
- **Trap:** Gleason 6 volume increase → treat — **Actually:** investigate (MRI, biomarker)
- **Trap:** REDEEM proves dutasteride prevents progression — **Actually:** volume cytorreduction; no upgrading difference
- **Trap:** negative MRI excludes progression — **Actually:** NPV depends on risk
- **Trap:** genomics for every GG1 — **Actually:** low prior → false positives ≈3× true risk
- **Trap:** stable PSA = stable disease — **Actually:** neuroendocrine change if hypogonadal